

# RETREIVE

Developer Build Brief | Version 1.0 | Confidential

## WHAT WE ARE BUILDING

A web app for MCAT students. They upload their study PDFs, read them aloud, and the app tracks every word they speak in real time. Text highlights word by word as they speak. Progress only advances when they are actually speaking. After each section, MCQ questions pop up to test retention. Basic analytics show how much they read and how well they scored.

## TARGET USER

MCAT retakers in the US and Canada preparing for medical school. They already have study PDFs. They upload them. We help them retain better. They spend \$2000 to \$5000 on prep. \$5 is nothing to them.

## CORE USER FLOW

- 01 User lands on `retreive.app`
- 02 Sees a rotating science fact card about why reading aloud improves retention
- 03 Clicks to sign up and pays \$5 via Stripe
- 04 Account created via Firebase Auth
- 05 User uploads their PDF from their device
- 06 PDF is parsed and text is extracted and cleaned
- 07 User reads text aloud in karaoke mode — Deepgram tracks every spoken word, text highlights in real time
- 08 Progress cylinder drains as they read — progress only advances when speaking
- 09 After each passage MCQ pop-up appears — green if correct with nudge, white if wrong with correct answer
- 10 Session progress saves to Firestore after every paragraph — never lost on refresh
- 11 Basic analytics shown at end — words read, accuracy, time spent

## TECH STACK

| LAYER              | TOOL             | NOTES                                                                          |
|--------------------|------------------|--------------------------------------------------------------------------------|
| Frontend           | React            | Single page app                                                                |
| Speech Recognition | Deepgram API     | Free \$200 credit on signup at <a href="https://deepgram.com">deepgram.com</a> |
| Authentication     | Firebase Auth    | Email based login                                                              |
| Database           | Firestore        | Real time writes after every paragraph                                         |
| File Storage       | Firebase Storage | User PDF uploads                                                               |
| Payment            | Stripe           | \$5 one time fee, individual account, PAN + bank account needed                |

|         |        |                               |
|---------|--------|-------------------------------|
| Hosting | Vercel | Free tier, deploy from GitHub |
|---------|--------|-------------------------------|

## DATABASE STRUCTURE — FIRESTORE

```
users/{userId}
name: string
email: string
createdAt: timestamp
hasPaid: boolean
```

```
sessions/{sessionId}
userId: string
pdfName: string
wordsRead: number
accuracy: number
timeSpent: number
progressPercent: number
paragraphsCompleted: array
```

```
mcqs/{mcqId}
topic: string
question: string
options: array
correctAnswer: string
correctNudge: string
wrongNudge: string
```

## OPENING SCIENCE CARDS

Ten cards shown randomly on every app open before session starts. Each card: bold uppercase title, regular body text, one emoji centered at the very bottom with padding. Dark background. Tap anywhere to skip. Cards rotate — no repeat until all ten shown. Store in Firestore as: id, title, body, emoji.

## PROGRESS CYLINDER

Vertical cylinder on the side of the screen. Green color. Drains from full to empty as user reads. Tied to word count progress. Smooth CSS animation. MCQs trigger at natural section breaks not percentage points.

## CRITICAL REQUIREMENT

**SESSION PERSISTENCE** — Every paragraph completion must write to Firestore IMMEDIATELY. Not on session end. Not on page unload. IMMEDIATELY after each paragraph. User must return to exact position after closing the app. This is non-negotiable. If this breaks, the product fails.

## MCQ POP-UP BEHAVIOUR

Appears as a modal after each passage. Two MCQs per passage. If correct — full screen green background, correct nudge text below answer. If wrong — full screen white background, correct answer highlighted, wrong nudge text explaining why. Tap to continue to next passage.

## NOT BUILDING IN V1

- No mobile app — web only, but mobile browser must be responsive
- No custom content library — user uploads their own PDF
- No social or community features
- No complex analytics dashboard — just basic numbers at session end
- No admin panel — Firestore managed directly for now

## BUILD ORDER

|        |                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------|
| WEEK 1 | Opening card screen, Firebase Auth setup, Stripe \$5 payment gate, PDF upload and text extraction                    |
| WEEK 2 | Karaoke engine with Deepgram speech recognition, progress cylinder, real time session persistence to Firestore       |
| WEEK 3 | MCQ pop-up flow with green and white screens and nudges, basic analytics screen                                      |
| WEEK 4 | Testing, bug fixes, deploy to <a href="https://retreive.vercel.app">retreive.vercel.app</a> , share with first users |

## DESIGN DIRECTION

Dark background throughout. Minimal and clean. No clutter. Large readable text for karaoke passages. One cosmic emoji per opening card centered at bottom. Green progress cylinder. White text on dark background. Fully mobile responsive.

## REPO AND ACCESS

GitHub repo owned by Phalesh. Developer gets collaborator access with write permissions only — not owner access. Firebase credentials and Stripe keys stay with Phalesh only. Developer works on frontend and integration. Never commits API keys to the repo.